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# A GPU-Accelerated Urban Flood Simulation Framework Integrating Green Infrastructure and Drainage Constraints

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## Abstract

This study presents a computationally efficient, GPU-accelerated pluvial flood simulation framework tailored for urban-scale deployment. The model integrates non-uniform rainfall input using Ordinary Kriging, hydrological runoff estimation via the SCS Curve Number method enhanced with a Green Infrastructure (GI) calculator, and sewer capacity constraints based on empirical catch basin density. A dynamic Weighted Cellular Automata (WCA2D) engine governs surface water redistribution, with spatially varying time steps derived from Manning's equation to improve both numerical stability and runtime efficiency. The model operates at a 10-foot ( $\sim 3$  m) resolution and completes a full 26-hour Hurricane Ida simulation in under two hours, achieving a 2 $\times$  speed-up over CPU-based implementations. A data-driven aggregation algorithm consolidates over 100,000 micro-basins into 135 hydrologically coherent divisions across New York City's 784 km<sup>2</sup> landscape. Validation against USGS flood reports and NYC's official stormwater maps yields a true positive rate of 77.5% and a 20% reduction in false alarms due to the introduction of a drainage capacity cap. Rainfall interpolation achieves a cross-validated RMSE of 2.26 inches and  $R^2$  of 0.12 across 29 stations. The resulting platform not only supports accurate high-resolution flood forecasting but also enables rapid scenario testing for GI retrofitting and stormwater infrastructure planning, providing actionable insights for urban flood resilience.

## 1. Introduction

The increasing impacts of climate change have intensified the risks of floods in urban areas, leading to significant economic and social losses. Among natural disasters, flooding results in some of the largest damages ([Padmaja et al., 2022](#)). Pluvial flooding occurs when heavy rainfall is too much for the ground to absorb or drainage systems to handle, causing water to accumulate on the surface. With a year-round humid climate, a large proportion of impervious surfaces due to high urbanization, and an old sewer system, NYC is at high risk of pluvial flooding. In addition, climate change is making it rain more intensely because warmer air holds more moisture. [Ortiz et al. \(2024\)](#) highlight that annual average rainfall in New York City has increased since 1960, emphasize the growing urgency for effective flood mitigation measures.

Flood simulation is a crucial tool for evaluating the damage caused by disasters and for designing effective protective measures. Accurate modeling of flood events facilitates risk assessment, prediction, and informed decision-making in urban and regional planning. Existing approaches to flood simulation can be broadly classified into hydraulic-hydrologic models and conceptual flood models, each offering specific advantages and limitations.

Hydraulic-hydrologic models are physically based approaches that integrate hydraulic processes, such as sewer network behaviors, with hydrologic processes governing surface water flow. These models provide precise risk assessments and predictions by capturing detailed interactions between drainage infrastructure and water movement ([Paniconi & Putti, 2015](#)). However, the primary limitations of hydraulic-hydrologic models include their reliance on detailed drainage data, substantial computational requirements, and high implementation costs. These constraints make their use challenging for large-scale applications or in scenarios with limited data availability.

In contrast, conceptual flood models simplify the representation of flood dynamics through basic assumptions about water flow and terrain. The Rapid Flood Spreading Method (RFSM), for example, defines flood zones based on wa-

ter volume-area relationships and the connectivity between zones [Nobre et al., 2011](#). Planar models simulate flood processes using water volume calculations and high-resolution digital terrain data to approximate inundation extents. Another approach, Cellular Automata (CA), models water flow using simple local rules that enable fast and parallel computations. The simplified nature of conceptual flood models makes them computationally efficient and suitable for large-scale floodplain analysis or rapid risk assessment. These models strike a balance between computational cost and usability while retaining sufficient accuracy for practical applications ([Teng et al., 2017](#)).

Among conceptual flood models, Cellular Automata (CA) approaches have gained prominence for their simplicity and computational efficiency. In these models, water flow is governed by local rules based on terrain slope, allowing water to move from higher- to lower-elevation neighboring cells at each time step ([Ghimire et al., 2012](#)). This structure enables fast flood simulations and is highly suitable for parallel processing. However, early CA models have notable limitations: they often ignore sewer network contributions, lack representations of hydrological losses such as infiltration, and their computation time increases significantly with domain size. To address some of these challenges, Weighted Cellular Automata (WCA) models were developed, which allocate water volumes to adjacent cells based on proportional weights and leverage GPU acceleration ([Guidolin et al., 2016](#)). WCA models are less sensitive to time-step size and benefit from multi-core or GPU hardware, though they can become sensitive to grid resolution and lose computational efficiency at finer scales. Despite these limitations, both CA and WCA approaches are widely adopted in urban flood modeling due to their balance between speed and spatial representation, making them well-suited for large-scale or rapid flood assessment tasks.

This study develops a computationally efficient, GPU-accelerated flood simulation framework tailored to rainfall-induced urban flooding. Building upon a conceptual structure, the model defines hydrological divisions based on high-resolution DEM data and simulates water movement across these units in response to spatially variable rainfall. The workflow integrates multiple components: (1) spatial interpolation of rainfall using kriging to capture non-uniform precipitation patterns; (2) runoff estimation through land use-based methods, including TR-55 and a Green Infrastructure (GI) calculator to quantify hydrological losses such as infiltration and retention; (3) drainage discharge estimation using available catchment bin data to approximate sewer system capacity under limited information; and (4) flood water transfer simulated using a Weighted Cellular Automata 2D model (WCA2D), which computes water redistribution based on local velocities de-

rived from Manning's equation. This integrated approach allows for the estimation of flood depth at specific locations, accounting for stormwater infrastructure and land surface characteristics. Its modular design and parallel computing compatibility make it suitable for large-scale, rapid flood impact assessments.

To verify and validate the proposed flood simulation methodology, model outputs are compared against spatial flood extents from the NYC Stormwater Flood Maps<sup>1</sup> and observed flood records from the United States Geological Survey (USGS)<sup>2</sup> during the 2021 Hurricane Ida remnants. This event, marked by widespread pluvial flooding and significant infrastructure disruption across New York City, serves as a valuable benchmark for both validating the model's spatial accuracy and verifying its responsiveness to extreme urban storm conditions. By balancing computational efficiency with practical reliability, this study contributes to the development of a scalable flood assessment tool suitable for urban environments increasingly threatened by high-intensity rainfall events.

## 2. Methodology

The proposed flood simulation framework consists of three modular components: terrain preprocessing, hydrologic bookkeeping, and a GPU-accelerated flow kernel. In the terrain preprocessing stage, hydrological basins are delineated from a high-resolution Digital Elevation Model (DEM) using the ArcGIS Basin tool, and adjacent basins are merged into compact hydraulic divisions through a graph-based Depth-First Search (DFS) algorithm, with the assumption of no cross-boundary water exchange. Hydrologic bookkeeping integrates rainfall and land cover data to estimate runoff using the SCS Curve Number method, while a Green Infrastructure (GI) calculator adjusts runoff volumes based on the presence of interventions such as rain gardens or permeable pavement. Sewer drainage capacity is estimated based on NYC's infrastructure design standard (1.75 inches/hour), with capacity allocated proportionally across divisions according to catch basin count and area. Flow velocity is computed using the Manning equation, where slope is derived from elevation gradients using a 3×3 neighborhood filter. Finally, water redistribution is modeled using a Weighted Cellular Automata (WCA) approach that enables efficient parallel computation on GPUs

<sup>1</sup>New York City Department of Environmental Protection. NYC Stormwater Flood Maps – Frequently Asked Questions. Retrieved from <https://www.nyc.gov/assets/dep/downloads/pdf/environment/education/stormwater-flood-map/swfm-faq.pdf>

<sup>2</sup>U.S. Geological Survey. (2021). USGS Responds to Hurricane Ida. Retrieved from <https://www.usgs.gov/news/national-news-release/usgs-responds-hurricane-ida>

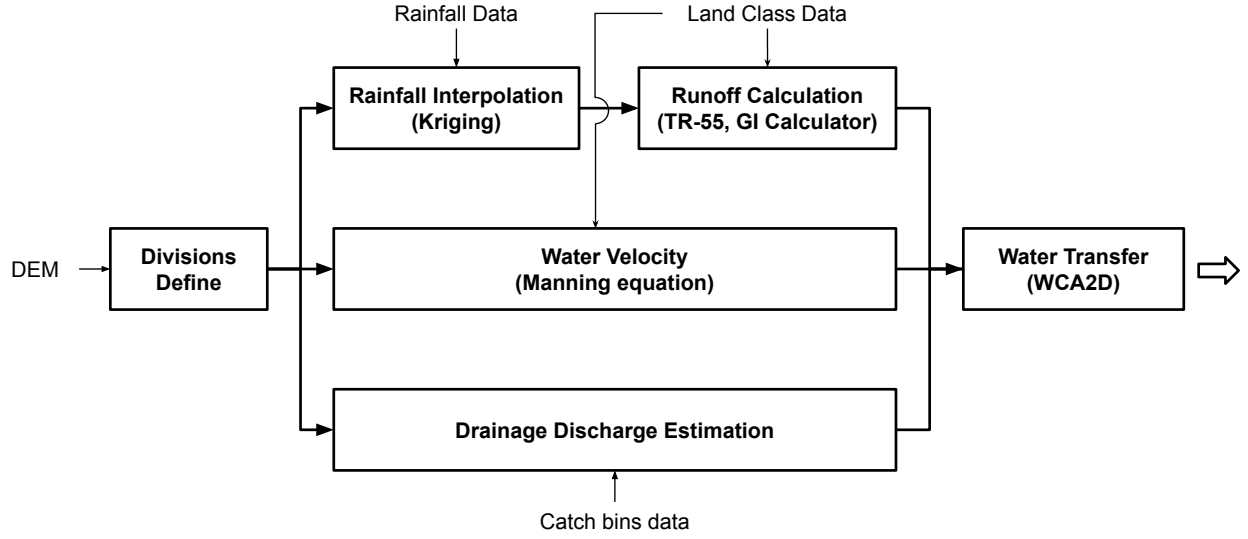


Figure 1. The proposed flood simulation framework consists of three modular components: terrain preprocessing, hydrologic bookkeeping, and a GPU-accelerated flow kernel.

by transferring flow volumes to neighboring cells based on relative elevation without the need for sorting operations.

## 2.1. Generate Divisions

Basin delineation is a fundamental process in hydrological modeling, as it divides a geographic area into distinct regions based on natural topography. These regions, or basins, represent areas where surface runoff is directed toward a common outlet, such as rivers, streams, or drainage points. By generating divisions based on natural topographic boundaries, it can be assumed that there is no significant water exchange between adjacent divisions. This assumption simplifies flood simulation by allowing localized water balance calculations within each basin, reducing computational complexity while maintaining accuracy. To reduce fragmentation and enhance computational efficiency, a graph-based refinement process is applied following initial basin delineation. Hydrological basins are first extracted from a Digital Elevation Model (DEM) using the ArcGIS Pro Basin tool. Due to DEM limitations, particularly at the study area boundaries, the output often includes small, fragmented basins. To address this, adjacent basins sharing a lower boundary are merged based on topographic heuristics. The spatial relationships between basins are then translated into a graph structure: nodes represent basins characterized by area and average boundary

elevation, while edges represent shared boundaries annotated with average elevation. Two pruning rules are applied to simplify the graph: (1) edges are removed if both basins exceed an area threshold, and (2) edges are excluded if their shared boundary is higher than both adjacent basins. A Depth-First Search (DFS) algorithm is then used to aggregate basins into larger hydraulic divisions. Starting from the smallest unvisited basin, DFS recursively merges connected basins until a predefined area threshold is met. This approach produces topographically coherent and computationally tractable divisions, suitable for efficient flood simulation.

## 2.2. Hydrologic Modeling and Runoff Estimation

### 2.2.1. RAINFALL INTERPOLATION

Rainfall intensity data required for the runoff calculations are interpolated using the Kriging method. Kriging is a geostatistical interpolation technique that accounts for spatial autocorrelation, assigning higher weights to nearby observations to produce an unbiased and minimum-variance estimate (Isaaks & Srivastava, 1989). The rainfall intensity  $i$  for each division is extracted from the interpolated surface, ensuring spatial consistency in input precipitation. The accuracy of the interpolation is evaluated using the Root Mean Square Error (RMSE) between observed and

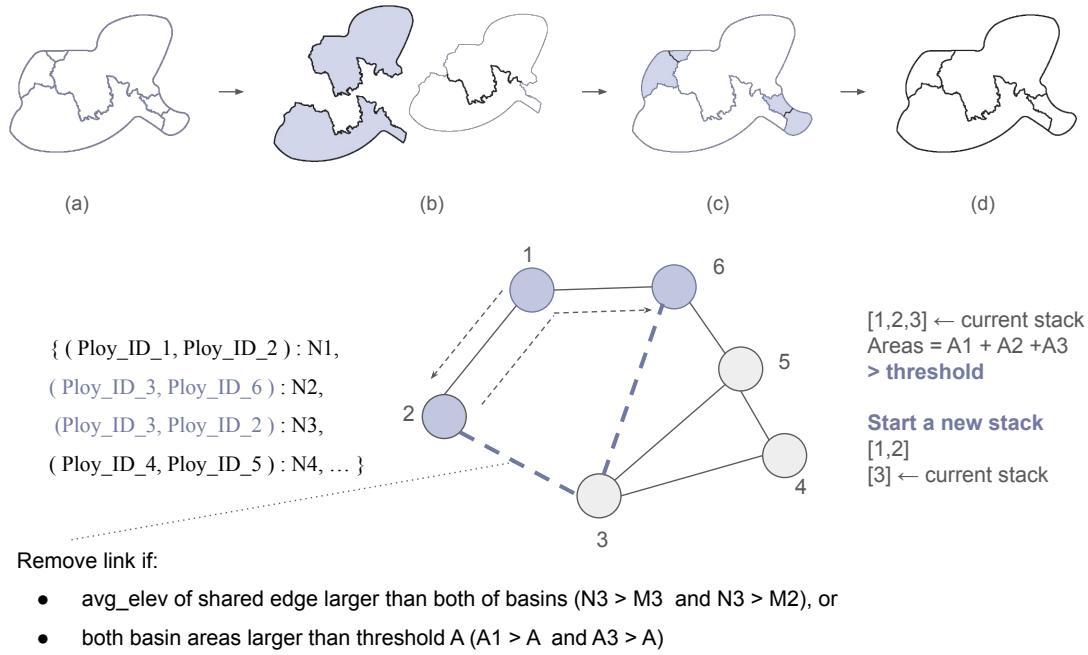


Figure 2. **Generate Divisions.** (a) Basin delineation by ArcGIS Pro basin tool. (b) Intercept and Zoning Statistic. (c) Depth First Search Algorithm. (d) Merge the Division.

estimated values, which serves as the primary criterion for selecting the optimal Kriging parameters.

### 2.2.2. RUNOFF CALCULATION

The SCS Runoff Curve Number Method, developed by the USDA Soil Conservation Service, estimates direct runoff from rainfall events (Hjelmfelt, 1991). It is based on the empirical relationship between rainfall, runoff, and watershed characteristics. The method consolidates initial abstractions—such as interception, depression storage, and infiltration—into a single term,  $I_a$ , representing the initial abstraction. Runoff begins only after rainfall exceeds this initial abstraction.

The SCS runoff equation is given by:

$$Q = \frac{(P - I_a)^2}{P - I_a + S}$$

Where  $Q$  is the runoff,  $P$  is the rainfall,  $I_a$  is the initial abstraction (losses like infiltration and surface storage), and  $S$  is the maximum retention. All variables are measured in inches. This equation models the relationship between rainfall, storage, and runoff for hydrological analysis.

Empirical analysis suggests that  $I_a$  is approximately 20% of the potential maximum retention,  $S$ , which is related to

the Curve Number ( $CN$ ) by:

$$S = \frac{1000}{CN} - 10$$

The key factors influencing the  $CN$  include the hydrologic soil group (HSG), land cover type, treatment, hydrologic condition, and antecedent runoff condition (ARC). A detailed procedure for selecting  $CN$  is outlined in (United States Department of Agriculture, 1986).

Flooding occurs when heavy rainfall exceeds the ground's absorption capacity or the drainage system's handling capacity, leading to surface water accumulation. To incorporate this effect into flooding simulation, we introduce the concept of maximum runoff ( $D_{max}$ ), which represents the drainage system's capacity. The final flood height equation is presented as:

$$F = Q - D_{max}$$

### 2.2.3. GREEN INFRASTRUCTURE CALCULATOR

Green Infrastructure refers to nature-based systems designed to manage stormwater sustainably by mimicking natural hydrological processes. The methodology classifies GI into three spatial types: point-based (e.g., rain gardens, tree pits), line-based (e.g., bioswales, green streets), and area-based (e.g., green roofs, permeable pavements,

and open space preservation). These interventions mitigate flood risk primarily through two hydrological mechanisms: by increasing the initial abstraction ( $I_a$ )—the volume of precipitation captured before runoff begins—and by reducing the runoff coefficient ( $S$ ), which represents the potential maximum retention. Together, these effects lower the peak runoff and delay stormwater discharge, reducing strain on urban drainage systems.

To incorporate the hydrological benefits of Green Infrastructure (GI) into the runoff estimation framework, we adjust the maximum potential retention storage ( $S$ ) in the TR-55 method based on GI implementation. The modified storage potential, denoted as  $S_{\text{with gi}}$ , accounts for both the volume retained by GI features and the reduction in impervious area. The adjustment is computed using the following expression:

$$S_{\text{with gi}} = \frac{V_{\text{with gi}}}{A_{lc}} = \frac{S_{lc} \cdot (A_{lc} - A_{gi}) + V_{gi}}{A_{lc}}$$

where  $S_{lc}$  is the original storage capacity for the land cover type,  $A_{lc}$  is the total area of the land cover,  $A_{gi}$  is the area occupied by GI interventions, and  $V_{gi}$  is the volume retained by the GI practice. This formulation, adapted from the Green Values Calculator methodology, reflects a weighted average of storage capacity by integrating the enhanced infiltration potential of GI systems (Center for Neighborhood Technology, 2022). The resulting  $S_{\text{with gi}}$  is then used within the TR-55 runoff model to produce a more realistic estimate of runoff under GI-enhanced conditions. By modifying  $S$  at the land-cover level, this method allows scalable and spatially explicit integration of GI performance into urban flood simulations.

The methodology also includes a generalized cost framework for evaluating GI strategies. Costs are calculated based on unit construction cost, maintenance frequency and cost, expected lifespan, and area coverage. Capital costs differ by GI type and urban context, while operation and maintenance costs are often modeled annually or as percentage-based estimates over the GI's lifecycle.

The Green Values Calculator enables users to input parameters such as site area, GI types and extents, rainfall scenarios, soil characteristics, and land cover conditions. It outputs a range of performance metrics including reduction in runoff volume, peak flow delay, pollutant removal, and lifecycle costs. These outputs support two primary applications: (1) improving the accuracy of flood simulation models by adjusting hydrologic parameters to reflect GI impacts, and (2) guiding urban planners and policymakers in optimizing GI deployment by comparing hydrological performance and cost-efficiency across different GI scenarios. This dual functionality makes it a valuable tool for

both simulation-based flood management and strategic GI investment planning.

#### 2.2.4. WATER VELOCITY

To estimate overland flow velocity across the simulation domain, we apply the Manning equation, a widely used empirical formulation that relates flow velocity to surface roughness, hydraulic radius, and slope. For each hydraulic division, the average flow velocity  $v$  is computed as:

$$v = \frac{1.49}{n} R^{2/3} S^{1/2}$$

where  $n$  is the Manning roughness coefficient (weighted by land cover),  $R$  is the hydraulic radius—typically approximated as the sheet flow depth—and  $S$  is the average terrain slope. Slope is calculated from the digital elevation model using a  $3 \times 3$  Sobel filter to estimate the terrain gradient components  $\frac{dZ}{dx}$  and  $\frac{dZ}{dy}$ , with the resulting slope magnitude computed as:

$$S = \sqrt{\left(\frac{dZ}{dx}\right)^2 + \left(\frac{dZ}{dy}\right)^2}$$

These average velocities are then used to define spatially varying time steps for floodwater redistribution. Specifically, the time step  $\Delta t$  for each division is determined by its local average velocity to ensure that the numerical fluxes are integrated with sufficient stability and resolution. This approach avoids the inefficiency of using a global time step constrained by the most restrictive cell, which often results in underutilized computational capacity across much of the domain. By tailoring time steps to local flow conditions, this method improves simulation performance and numerical accuracy, in line with findings from previous studies on adaptive time stepping in hydraulic models (Zhang & Cundy, 1994; Wright & Hedges, 1999; Crossley & Wright, 2003).

#### 2.2.5. DRAINAGE DISCHARGE ESTIMATION

The drainage capacity  $D_i$  for each division  $i$  is derived based on the total design capacity of New York City's sewer system, which is assumed to manage 1.75 inches of rainfall per hour. The discharge for each division is distributed proportionally according to its area and the number of catch basins present. The maximum discharge capacity for division  $i$  is expressed as:

$$D_i = \frac{(1.75 \cdot A)}{N} \cdot \frac{n_i}{A_i}$$

Here,  $A$  is the total area of New York City,  $N$  is the total number of catch basins in the city,  $n_i$  is the number of

catch basins in division  $i$ , and  $A_i$  is the area of division  $i$ . This equation ensures that the discharge capacity is allocated based on both spatial distribution and infrastructure density.

### 2.3. Inter-cellular-Volume Computation for Water Redistribution

To simulate surface water flow across cells, we implement an inter-cellular-volume redistribution scheme based on weighted slope differences and local water surface elevation. At each time step, the water surface elevation at each grid cell  $(i, j)$  is calculated as the sum of the digital elevation model (DEM) elevation  $z_{i,j}$  and the water depth  $w_{i,j}$ :

$$h_{i,j} = z_{i,j} + w_{i,j}$$

Each cell then compares its water surface elevation to that of its adjacent neighbors. The slope difference to a neighboring cell located at offset  $(\Delta r, \Delta c)$  is computed as:

$$\Delta h = h_{i,j} - h_{i+\Delta r, j+\Delta c}$$

Flow is permitted only if  $\Delta h$  exceeds a predefined slope threshold  $\epsilon$ , which filters out negligible differences and improves computational efficiency. The total volume of water to be redistributed from cell  $(i, j)$  is defined as a fraction  $\alpha$  of its current water content:

$$w_{\text{out}}^{(i,j)} = \alpha \cdot w_{i,j}$$

This outgoing volume is then proportionally distributed to eligible downslope neighbors according to their relative slope differences:

$$w_{\text{flow}}^{(i+\Delta r, j+\Delta c)} = \frac{\Delta h}{\sum \Delta h} \cdot w_{\text{out}}^{(i,j)}$$

This method ensures that water flows preferentially toward steeper descent directions while maintaining mass conservation. The redistribution is repeated iteratively over time steps to simulate continuous surface water movement. Due to its local and parallel structure, the scheme is well suited for GPU-accelerated flood modeling based on weighted cellular automata frameworks.

## 3. Result

### 3.1. Study Area

The study area encompasses all five boroughs of New York City, covering a total land area of approximately 783.8 million square meters. The simulation is conducted at a spatial



Figure 3. Study area of New York City used in the flood simulation. The map shows 135 hydrological divisions derived from basin merging, overlaid on a digital elevation model (DEM). Masked areas (in light gray) were excluded from the simulation due to insufficient drainage data, including large parks, airports, and industrial parcels.

resolution of approximately 3 meters, based on a raster grid of size  $14,925 \times 15,128$  with 10-foot ( $\sim 3$  m) cells.

Flood-prone zones are identified as areas where modeled water depth exceeds 3 meters. Certain regions are excluded from analysis due to insufficient information on drainage infrastructure. These masked-out zones include large open space parks (areas  $\geq 100,000$  ft<sup>2</sup> from the NYC Planimetric Database), industrial or non-commercial PLUTO parcels  $\geq 250,000$  ft<sup>2</sup>, and parcels intersecting railway infrastructure or airports.

To support scalable hydrological simulation, the city is divided into 135 hydrological divisions. These are derived by aggregating over 100 Southlands basins generated using the ArcGIS Pro Basin tool. Basin merging is performed using a graph-based approach under area constraints. On average, each division covers approximately 53.45 million ft<sup>2</sup> (around 5 million m<sup>2</sup>), allowing for spatially resolved simulation of flood dynamics across the urban terrain.

In addition to topographic and hydrological delineation, infrastructure and surface characteristic data are integrated to enhance accuracy. Catch basin data were obtained from the NYC Department of Environmental Protection (DEP) open dataset, representing stormwater collection points distributed throughout the five boroughs. Each hydrological division is associated with the number of catch basins it contains, which is used to approximate localized sewer drainage capacity in the absence of complete pipe network data. Furthermore, surface runoff and flow resistance are modeled using land cover classifications derived from DEP's Citywide Parcel-Based Impervious Area (LCpp) dataset. Each land parcel is assigned a land cover

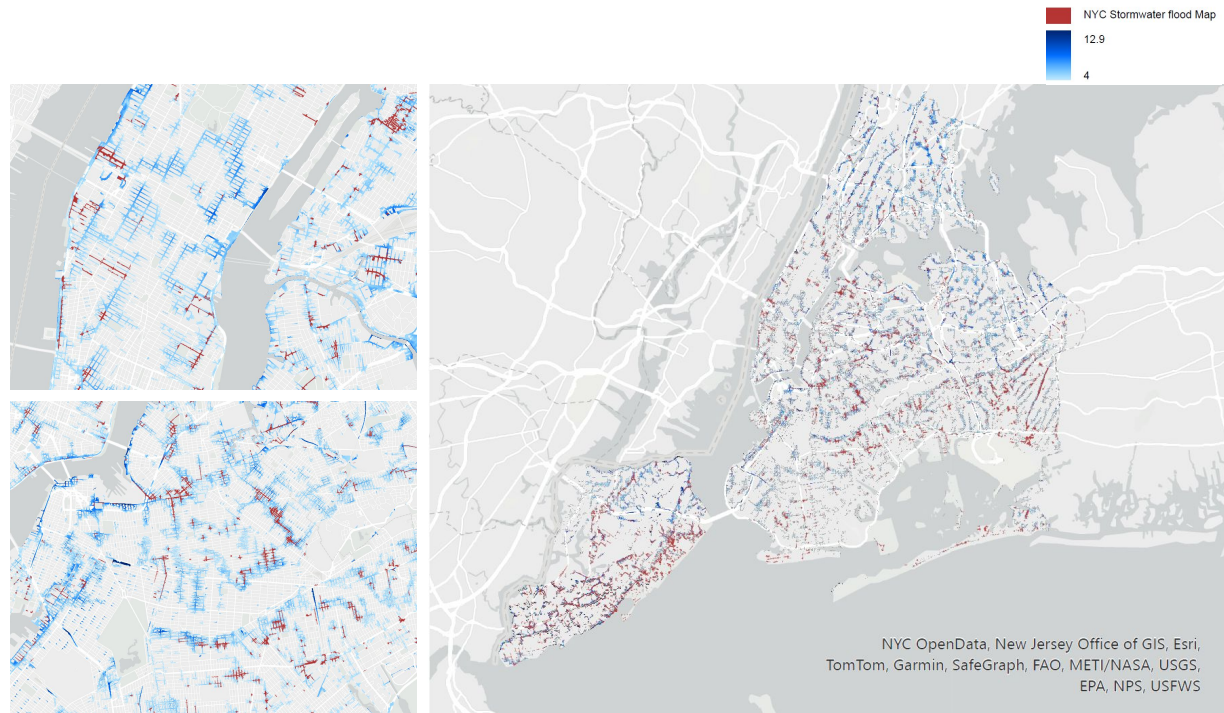


Figure 4. Comparison of simulated flood extents with the NYC Stormwater Flood Map for a 2.13-inch rainfall scenario. Blue areas indicate predicted water depth from the proposed flood simulation model, while red overlays represent reference inundation areas derived from the NYC Stormwater Flood Map. Insets highlight spatial agreement and discrepancy in dense urban contexts such as Manhattan and Brooklyn.

type (e.g., asphalt, roof, grass), which informs both the runoff coefficient (C-value) for use in TR-55 runoff calculations and the Manning’s roughness coefficient ( $n$ ) for hydraulic flow estimation. The table 1 summarizes the corresponding hydrologic and hydraulic parameters associated with major land cover types.

Table 1. Land cover classes and their associated runoff coefficients (C-values) and Manning’s roughness values ( $n$ ). C-values are based on NYC DEP parcel-based impervious area classifications. Manning’s  $n$  values are referenced from (Chow, 1959; Natural Resources Conservation Service, 2004; ?) for common urban surfaces.

| Land Cover     | Runoff Coefficient | Manning’s $n$ |
|----------------|--------------------|---------------|
| Asphalt        | 0.98               | 0.011–0.013   |
| Concrete       | 0.85–0.98          | 0.012–0.015   |
| Roof           | 0.85–0.95          | –             |
| Brick Paver    | 0.80–0.98          | 0.012–0.015   |
| Synthetic Turf | 0.25–0.70          | 0.030–0.035   |
| Wood           | 0.85–0.98          | 0.015–0.018   |
| Grass          | 0.10–0.35          | 0.10–0.15     |
| Bush           | 0.10–0.35          | 0.20–0.40     |
| Tree           | N/A                | 0.30–0.50     |
| Bare Soil      | 0.15–0.50          | 0.02–0.05     |
| Sand           | 0.30–0.50          | 0.02–0.04     |
| Gravel         | 0.25–0.85          | 0.02–0.035    |
| Open Water     | N/A                | 0.025–0.035   |

### 3.2. Moderate Flood (2.13 inches/hr)

To validate the simulation results, we compare the predicted inundation extents with the Moderate Flood (2.13 inches/hr) with Current Sea Levels scenario from the NYC Stormwater Flood Maps, which represents a typical 10-year storm with approximately a 10% annual probability of occurrence. This citywide model simulates peak ponding depth under the scenario and outputs 2D flood layers processed through GIS. The raw model results are smoothed and aggregated to highlight only areas of contiguous flooding exceeding 0.25 acres in size and a minimum ponding depth of 4 inches. Flood depths are categorized into two layers: depths greater than 4 inches, and those exceeding 1 foot. The study use the same post-processing procedure and the total runtime for 1 hour rainfall simulation is 24min 46s.

The TPR and FDR metrics, also respectively called probability of detection (hit rate) and false alarm ratio (Bennett et al., 2013), are used to compare the inundated area predicted by the two models. we evaluate our model performance with a true positive rate (TPR) of 0.7748 and a false discovery rate (FDR) of 0.1946. The overestimation in inundated extent is attributed to masked or missing drainage infrastructure information in the reference dataset, particularly in regions where catch basin or sewer data is incomplete.

To improve both computational efficiency and predictive accuracy, we implemented a spatially varying time step derived from Manning’s equation, allowing the model to dynamically adjust the temporal resolution based on local terrain and flow conditions. This approach significantly reduced runtime while maintaining high detection performance. As shown in Figure 5, the Manning-based substepping achieved a true positive rate (TPR) of 0.777 at a shorter runtime compared to numerically integrated fixed-step methods. In parallel, we introduced an empirical drainage limit parameter ( $d_{\max}$ ) that constrains maximum discharge per division, mitigating overprediction in flood extent. Figure 6 shows that increasing  $d_{\max}$  lowers the false discovery rate (FDR) while only slightly reducing TPR. Together, these enhancements strike a favorable balance between runtime efficiency, model sensitivity, and spatial accuracy in high-resolution flood simulation.

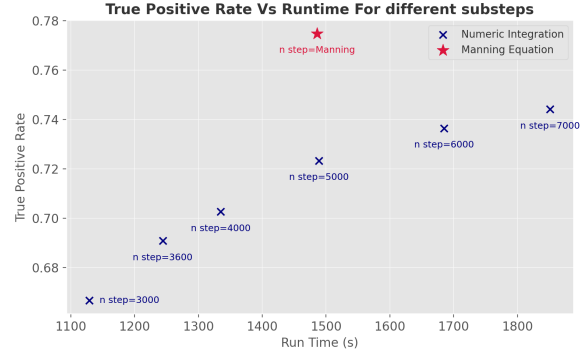


Figure 5. Tradeoff between true positive rate (TPR) and simulation runtime across different substep counts. The red star indicates the performance of the Manning-based spatially varying time step approach, which yields the highest accuracy with lower runtime.

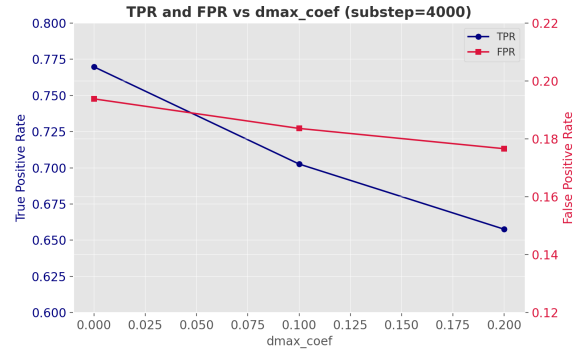


Figure 6. Sensitivity analysis of the maximum drainage limit coefficient ( $d_{\max}$ ). As  $d_{\max}$  increases, the false positive rate (FPR) decreases while TPR slightly declines, suggesting a controllable tradeoff between precision and recall.

### 3.3. Ida Remnants pluvial flooding

Table 2. Rainfall interpolation accuracy for the Hurricane Ida event (48-hour accumulation).

| Metric         | Ida Event        |
|----------------|------------------|
| # of Stations  | 29               |
| $R^2$          | 28.62%           |
| RMSE (48 hour) | <b>2.26 inch</b> |
| MAE (48 hour)  | <b>1.93 inch</b> |

To evaluate the accuracy and reliability of the proposed flood simulation methodology, the results of the simulation are compared with actual observed inundation data from Hurricane Ida remnants, which impacted New York City (NYC) on September 1, 2021. The remnants of Hurricane Ida produced unprecedented rainfall across the five boroughs of NYC, resulting in severe pluvial flooding. To validate the proposed flood simulation methodology, actual

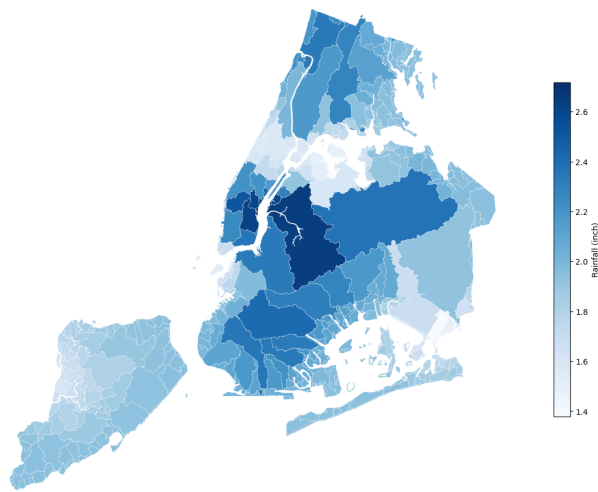


Figure 7. The spatial distribution of rainfall intensity across New York City (NYC) on September 1, 2021, at 23:00 during the remnants of Hurricane Ida. The rainfall data has been interpolated using Ordinary Kriging with a linear kernel, producing a continuous rainfall surface. The map clearly demonstrates significant spatial variation in rainfall intensity, with darker blue areas indicating higher rainfall values exceeding 4 inches and lighter areas representing lower rainfall values around 2.25 inches.

flood inundation reports were sourced from USGS and Local Storm Reports from National Weather Service.

Rainfall input for the simulation was generated using Ordinary Kriging with a linear variogram model to interpolate hourly rainfall intensity across the study domain. The resulting spatial distribution (Figure 7) reveals significant heterogeneity, with localized peaks exceeding 3.5 inches per hour—well above New York City’s sewer system design threshold of 1.75 inches per hour. The interpolation accuracy was evaluated through cross-validation using 29 rain gauge stations, yielding a coefficient of determination ( $R^2$ ) of 28.62% a root mean square error (RMSE) of 2.26 inches and a mean absolute error (MAE) of 1.93 inches over a 48-hour period (Table 2). These results reflect the challenges of capturing spatially concentrated rainfall extremes, particularly under limited gauge coverage during the Hurricane Ida event.

The model’s predictive accuracy was evaluated using 131 flood observation points reported by the USGS during Hurricane Ida. As shown in Figure 8 and summarized in Table 3, 67 of these points (51.1%) were located within the model-predicted flood zones, indicating reasonable spatial agreement between simulated and observed flood extents. The results are particularly aligned in historically vulnerable areas such as southeast Queens, southwest Brooklyn,

and lower Manhattan, where ponding and runoff were most intense. The simulation was completed in 1 hour and 40 minutes at a 10-foot (approximately 3-meter) spatial resolution, covering the full 26-hour duration of the Ida storm event. These findings demonstrate the model’s ability to resolve localized pluvial flooding patterns at city scale while maintaining operational efficiency. However, some false negatives remain, primarily in areas where drainage infrastructure data is incomplete or missing, highlighting opportunities for further refinement in future implementations.

Table 3. Summary of Hurricane Ida flood simulation validation using USGS-reported flood points.

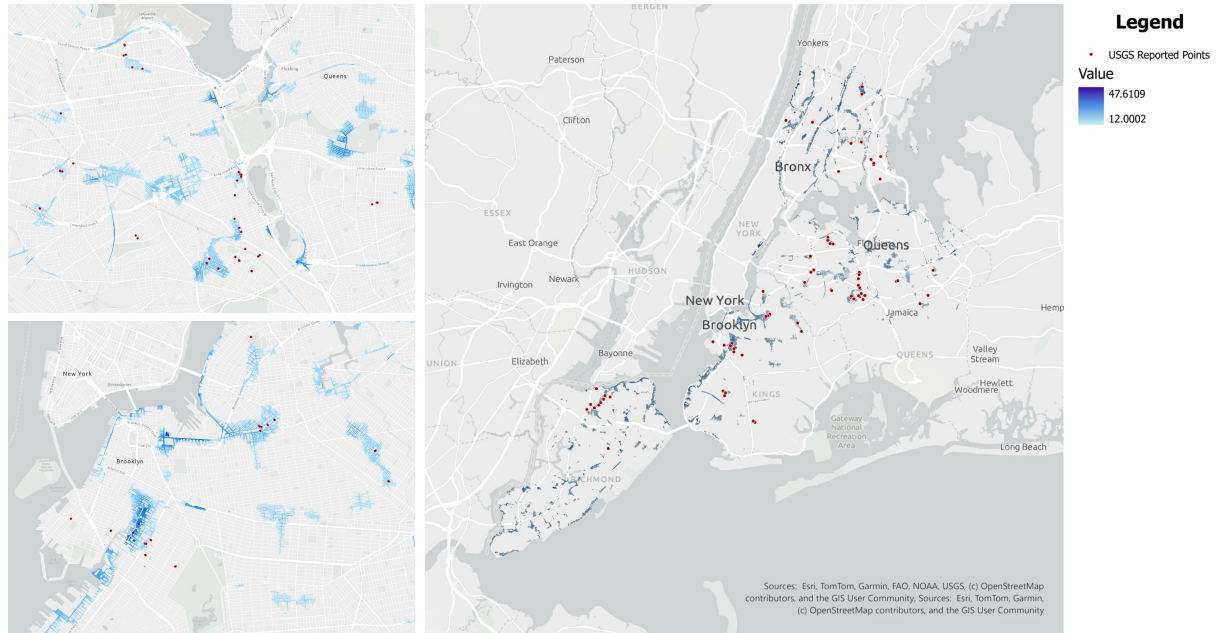
| Metric                           | Value               |
|----------------------------------|---------------------|
| USGS report points in study area | 131                 |
| Positive predicted               | 67 (51.1%)          |
| Total simulation runtime         | 1 hour 40 minutes   |
| Total storm duration (Ida event) | 26 hours            |
| Spatial resolution               | 10 feet (~3 meters) |

#### 4. Discussion and Conclusion

This study presents a scalable, GPU-accelerated rainfall-induced flood simulation framework that balances computational efficiency with hydrological realism, offering timely predictions at city scale. By integrating non-uniform rainfall inputs via kriging, our model captures spatially variable storm intensity—essential for dense urban environments like New York City. The validated rainfall interpolation (RMSE = 2.26 in over 48 hours) demonstrates reasonable alignment with 29 observed stations during Hurricane Ida, though challenges remain in representing localized extremes due to sparse gauge distribution. Nevertheless, the spatial rainfall structure supports physically plausible runoff generation and flow routing at a 3-meter resolution across the entire 784km<sup>2</sup> domain.

A core innovation of this platform is its integration of hydrological loss modeling using the SCS Curve Number method enhanced by a Green Infrastructure (GI) calculator. By modifying curve numbers and storage capacity based on GI type and extent, the model evaluates the mitigation potential of retrofitted landscapes, including rain gardens and permeable pavement. While the implementation enables scenario testing for GI deployment, further empirical validation is required to calibrate infiltration and retention effects under varying site conditions and storm profiles. The current parameterization is based on generalized literature estimates, which may not capture micro-scale hydraulic interactions in heterogeneous parcels.

Our sewer drainage capacity estimation introduces a novel, data-constrained approach that allocates flow thresholds to



**Figure 8.** Validation of simulated flood extents against USGS-reported flood locations during Hurricane Ida. Blue areas indicate model-predicted floodwater depths, while red points denote official USGS flood observation reports. Insets provide zoomed-in views of spatial agreement in Queens, Brooklyn, and lower Manhattan. The comparison demonstrates good alignment between simulated and observed flood locations, particularly in historically flood-prone neighborhoods.

each division based on known catch basin density. This proxy-based cap curtails unrealistic outflows in under-instrumented areas and significantly reduces model over-prediction. However, the method’s reliance on empirical assumptions introduces inherent uncertainty. While effective in lowering the false discovery rate (FDR) by 20%, it may underestimate or misrepresent drainage performance in specific neighborhoods where undocumented infrastructure (e.g., private drains or legacy sewers) plays a significant role. Additional data collection or integration with sewer network models would improve this component’s fidelity.

Finally, a graph-based basin aggregation algorithm collapses over 106,000 micro-basins into 135 hydraulically coherent divisions, preserving topographic realism while maintaining computational tractability. This division scheme ensures that flow dynamics are resolved within topographically bounded units, accelerating simulation while retaining localized accuracy. The full WCA2D model completes a 2.13-inch design storm simulation in under 2 minutes on a GPU—achieving a 2× speed-up over a CPU implementation and enabling rapid experimentation. The platform is thus well-positioned for operational applications, offering actionable insight into GI investment plan-

ning and stormwater system upgrades. Future work will expand validation efforts—particularly for GI retrofitting scenarios—and enhance drainage modeling using infrastructure datasets as they become available.

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